

Week 1.4 — First Frame Sketches — Motors, Frame, CAD

Sunday, July 5, 2026 · 3:00 — 5:30 PM

Class 1.4 was the day the ROV started to look like a machine, not a pile of parts. Eleven kids showed up — Trenton, Rhylan, Eli, Oliver, Luke, Frankie, Rebecca, Liam, Maddox, Evan, and Carter. Meeting ran 3:00 to 5:30. The night moved on three parallel tracks: the control board kept getting soldered (with a wrinkle we're still working through), the motors got propellers assembled onto them for the first time, and the team started sketching frame ideas on paper and in CAD. On the control board — the wider iron tip we swapped in last week is causing temperature-regulation issues. Coach John's call: order the original tips. Sometimes the right answer to a tool problem is not to work around it. On the motors — propellers went onto the motor shafts. We reviewed the Sabertooth motor controller and how a modulator and a potentiometer work in that circuit. On the frame — kids drew top / side / front / bottom views with a key marking PVC, tether, thrusters, and thrust direction. Then Coach John moved into CAD and started the first Onshape design. A tutorial goes out so students can self-learn Onshape between meetings.

11 STUDENTS	5 FRAME SKETCHES	1 ONSHAPE DESIGN STARTED	2 hr 30 min MEETING DURATION
WHAT WE LEARNED - When a workaround isn't working, name it and replace the tool. The wider tip solved 1.3's oxidation but introduced temperature instability. We reverted; we didn't paper over it. - Frame design starts on paper. Sketch top, side, front, and bottom views with a symbol key. Proportions get sanity-checked at the paper stage before any CAD gets drawn. - Motors + propellers + a controller = the ROV's drivetrain. Understanding the Sabertooth, the modulator, and the potentiometer lets you diagnose a bad run instead of guessing. - CAD is a skill the team owns individually, not just something Coach John does. Between-meeting self-study on Onshape is how every student gets fluent enough by Phase 2.		DECISIONS MADE - Order the original iron tips. The wider tip fixed 1.3's oxidation but introduced a temperature-regulation problem. Sometimes the answer is to un-do the previous fix. - Frame design starts on paper. Top / side / front / bottom with a symbol key, before any CAD is opened. - Onshape is the team's CAD tool. Between-meeting self-study is on the roster — a tutorial goes out this week. - Officer elections still deferred until the summer TriggerFish build is wrapped. Same rule as 1.3.	
STUDENT LEARNING OUTCOMES - First hardware-in-hand moment for the drivetrain — propellers seated on motor shafts, not on a slide. - First drawings of the robot: top, side, front, bottom views on paper, with the team's first shared symbol key (PVC, tether, thruster, cross-tether, thrust direction). - Introduction to Onshape — the CAD tool the team will use through Phase 2 and beyond. Self-study track begins this week. - A working mental model of the Sabertooth motor controller and how the modulator and potentiometer shape the signal to the motor.		ACTION ITEMS - Order the original iron tips for the soldering stations <i>Coach John · by 1.5</i> - Send the Onshape CAD tutorial to every student for self-study <i>Coach John · this week</i> - Continue the TriggerFish control-box build once the correct tips arrive <i>Sunday cohort · 1.5</i> - Refine the frame sketches into a shortlist to CAD next week <i>Frame group · by 1.5</i> - Photograph the paper sketches and upload to the team folder <i>Photographer · by 1.5</i>	

LOOKING AHEAD

Class 1.5 continues the TriggerFish build once the original tips arrive, and the frame group narrows the sketches to a shortlist to CAD in Onshape. Everyone should have watched the CAD tutorial by then.

